CHRISTO JOHN

FULL STACK SOFTWARE ENGINEER

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GitHub | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

Full Stack Software Engineer building web products and cloud-backed systems across Next.js/React, TypeScript, Python/Django, PostgreSQL/Supabase and AWS/GCP. Experience includes independent client delivery, product architecture, third-party API integration, payment and authentication workflows, distributed media processing, and data-intensive dashboards. Combines formal UI design experience with security-minded development, test gates, and controlled AI-assisted engineering.

TECHNICAL COMPETENCIES

Languages: TypeScript, JavaScript, Python, SQL; working familiarity with Java and C/C++

Frontend & Product: React, Next.js (App Router/Server Components), Tailwind CSS, shadcn/ui, Radix UI, Figma, responsive UI

Backend & Data: Django/DRF, Next.js server APIs, PostgreSQL, Supabase, RLS, REST APIs, webhooks, Celery/Redis exposure

Cloud & Infrastructure: AWS S3, SQS and Lambda; GCP Compute Engine/Cloud Run; Docker, Vercel, object storage

AI, Automation & Media: LLM integration, structured extraction, Codex, Claude Code, Playwright, Remotion, Lottie, FFmpeg, Make.com, n8n

Engineering Quality: Vitest, Testing Library, Playwright E2E, TypeScript, ESLint, SQL migrations, security testing, ADRs, PRDs

PROFESSIONAL EXPERIENCE

Full Stack Developer | Letters to Abroad · Freelance | Sep 2026–Present

- Own requirements research, technical planning, architectural decisions and client communication for software supporting international education and employment services; delivered four website pages.
- Built a Next.js job-discovery experience using the German Federal Employment Agency's live API, including domain mapping, bilingual user-interface presentation, stable pagination and resilient handling of inconsistent upstream data.
- Investigated undocumented API constraints—sorting instability, pagination limits, search ambiguities and malformed job descriptions—and translated findings into explicit data and UI contracts.
- Integrated AI-supported job information enrichment and clarification experiences in the broader client work; scoped additional CRM, employer/candidate and support workflows without representing them as completed launches.

Full Stack Engineer | Neuegrid Design Studio | Jan 2026–Present

- Take products from requirements and system design through implementation, technical verification and deployment; connect UI/UX decisions with application architecture and cloud operations.
- Developed a browser-driven URL-to-video processing system using Playwright, FFmpeg, Docker and Google Cloud Compute Engine, addressing browser capture and media-processing constraints.
- Worked on a cloud-backed video product combining Next.js, Supabase, Remotion and AWS S3/SQS/Lambda; handled asynchronous processing, resource constraints and deployment boundaries.
- Applied AI coding agents within repository-specific constraints covering architecture context, documented decisions, tests, type checks, builds and final validation—not as substitutes for engineering review.

PROFESSIONAL EXPERIENCE — CONTINUED

Full Stack Developer Intern | Tomatrix Technologies | Oct 2025–Jan 2026

- Developed features of EduFlick AI, an education platform built with Next.js and Supabase, spanning learner authentication, onboarding, registration, enrollment and administrative interfaces.
- Implemented and debugged Razorpay order/payment verification, callback handling and post-payment navigation; worked with Supabase PostgreSQL and role-based security policies.
- Integrated existing n8n/chat-assistant workflows and supported redesign and production-readiness fixes on the LMS and main company website.

UI Design Intern | Neuegrid Design Studio | May 2025–Aug 2025

- Created Figma interface designs and brand/social assets for client projects; applied usability, responsive-layout, hierarchy and prototyping principles later carried into full-stack development.

SELECTED ENGINEERING PROJECTS

Motion Studio — No-Code Programmatic Video Platform

Next.js 16 · React 19 · Supabase · Remotion/Lottie · AWS S3/SQS/Lambda · Dodo Payments

- Built a browser-based editor for composing template-driven video advertisements and exports, with persistent projects and render-job state managed through Supabase.
- Architected an asynchronous export path: client-side asset preparation, direct S3 presigned uploads, server-authorized debit, SQS queueing, AWS Lambda consumption, Remotion Lambda rendering and S3 output/thumbnail delivery.
- Implemented idempotent render-specific billing and compensating refunds for enqueue or rendering failures; kept server-side authorization authoritative rather than trusting browser prechecks.
- Separated web and worker deployments, introduced concurrency limits, handled terminal cleanup and documented operational contracts and recovery cases.

stocketREX — Financial Recommendation Research Dashboard

Next.js · TypeScript · Supabase/PostgreSQL · RLS · Zod · Vitest · Playwright

- Designed a private research dashboard over a recorded audit of 31,569 logical recommendation calls, with 29,215 records in the defined processing cohort; maintained distinct source and eligibility boundaries.
- Modeled open/close matching, direction-aware performance and risk/reward calculations, normalization, filtering and analytical drill-downs while accounting for inconsistent source semantics.
- Designed owner authorization, database RLS, read-only source boundaries, forward migrations and testable data contracts to prevent silent errors or unauthorized mutation.
- Maintained requirement-to-implementation traceability, architecture decisions and verification gates across unit, browser, type-check and build workflows; did not present experimental ML features as deployed.

URL-to-Video Engine — Automated Website Recording

Next.js · Playwright · FFmpeg · Docker · GCP Compute Engine

- Built a capture and recording pipeline that automated Chromium navigation and scripted website scroll sessions before rendering them into reusable video output.
- Worked through virtual-browser resource constraints, frame-capture reliability and containerized execution in a cloud VM environment.

SELECTED ENGINEERING PROJECTS — CONTINUED

Pinsta — Multi-Platform Publishing & Reconciliation Hub

Next.js · TypeScript · Supabase · Make.com · S3-compatible storage · HMAC webhooks

- Built a single-owner media management application for coordinating content across Instagram, Pinterest and LinkedIn with canonical feed metadata, signed uploads and provider-linked media references.
- Designed publication state machines that distinguish definitive failures from ambiguous external outcomes, requiring reconciliation before unsafe retries that could duplicate posts.
- Enforced server-side secrets, owner-only access, RLS and HMAC-authenticated callbacks; kept provider tokens out of the application and used fail-closed gates for unverified publishing capabilities.
- Supported owner-scoped resumable media uploads and storage abstraction across Supabase, S3-compatible storage and local persistence, without automatically copying imported provider media.

LTA Jobs — Live Government Job Data Application | Client system

Next.js 16 · React 19 · TypeScript · Vitest · Testing Library · REST API

- Separated UI, presenters, core models and external services in a one-directional architecture, preventing German upstream API fields from leaking into application components.
- Implemented server-rendered listings with streaming/Suspense, typed German/English UI text, stable pagination, empty/error states and mapping safeguards for malformed external records.
- Enforced architectural import boundaries and production checks; measured real API behavior where public documentation was incomplete.
- Separated source errors from actual not-found states, ensuring API outages do not appear as withdrawn vacancies and slow sections do not block the rest of the page.

Virtual Background Photobooth — Real-Time Browser Media | GitHub source

MediaPipe · WebGL · Browser camera APIs · Supabase

- Built a Chromium kiosk photobooth using webcam segmentation and real-time replacement with image or video backgrounds; supported captured-photo delivery via QR pairing.
- Implemented separate kiosk/mobile experiences and a Supabase relay for phone handoff and remote shutter control; preserved local camera/background functionality when the network was unavailable.

AI Website Scraping Engine — Structured Web Extraction

Apify · Headless browser · Gemini API · JSON Schema · structured outputs

- Built an Apify actor that renders JavaScript-heavy websites and extracts requested information using natural-language instructions, optional JSON Schema constraints and structured JSON output.
- Added translation options and indexed extraction artifacts; treated model-derived data as an integration needing verification rather than guaranteed factual output.

EDUCATION

Master of Computer Applications (MCA) | St. Joseph's College of Engineering and Technology, Palai | 2025–2027 (expected)

Bachelor of Computer Applications (BCA) | Mar Augusthinose College, Ramapuram | 2022–2025

LEADERSHIP & RECOGNITION

- Acsia Technologies in-company hackathon: top-20 finalist among 240+ teams; advanced to Round 4 of EY Global Techathon 6.0.
- Led the MuLearn campus chapter: helped grow membership from 7 to 144 and organized a web-development bootcamp attended by 177 participants.
- Captained MCA teams across four regional hackathons, coordinating rapid prototypes and team delivery.

EARLIER PUBLIC ENGINEERING WORK

MuCompanion — Firebase/Firestore learning-task dashboard integrating MuLearn community tasks and progress tracking.

Complaint Management System — PHP/MySQL workflow with authentication, submissions, case tracking and administrative views.